

# ADR 0118 — A new desk is named, not located, and it is built in a root the app owns

- **Status:** Accepted — path policy decided; implemented in #549, amended by review (see §3a, §4)
- **Date:** 2026-09-04
- **Issue:** #549 (M11.04), answering `docs/superpowers/specs/m3.23-worktrees.md` **open question 2**
- **Extends:** ADR 0117 (*"A discovered desk needs a door..."* — lands with #548 / PR #654, still open at the time of writing, so the link is deliberately omitted rather than left to 404) · [ADR 0008](#) (the managed root this mirrors)
- **Supersedes / superseded by:** —

## Context

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`AddWorktree` is the first operation in M11 that **creates a directory**. Every other worktree operation so far has read (`git worktree list`) or selected (`/api/select-worktree`). Creating one raises a question none of them did:

**Where does the new directory go, and who chooses the location?**

The spec left this open deliberately, and Tom answered it on the issue: **a managed root**, the way clones already work.

This ADR records that decision, the argument for the alternative — which was real, and is Tom's own working habit — and one thing the decision forces that the spec's own sketch got wrong.

## Decision

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### 1. New worktrees live under a root the app owns

`worktrees_root()` mirrors `clones_root()` exactly (ADR 0008): an env override, else `$XDG_DATA_HOME/git-vista/worktrees`, else

`~/.local/share/git-vista/worktrees`. It is created at startup, admitted to the allowed roots **once**, and scanned like the clones root is.

The property that decides it, in Tom's words on the issue:

A managed root is inside the fence **by construction**. A sibling directory has to be checked against the fence every time a path is picked, and "checked every time" is the kind of rule that holds until one code path forgets it.

## 2. The operation carries a name, not a path

The spec sketches:

```
AddWorktree { path: PathBuf, branch: BranchName },
```

**That shape cannot ship here, and the managed root is why.** If the client supplies a path, then "is it inside the managed root?" becomes a check — and the whole point of the managed root is to have no check to forget. It also contradicts an invariant this codebase already holds everywhere else: request bodies do not carry paths. `SelectRequest`'s own doc says so (*"like every request body this cannot carry a path"*), `WorktreePathsRequest` validates each element into a `WorktreePath` that can never be absolute and can never carry `..`, and `/api/delete-clone` addresses a directory by opaque id rather than by location.

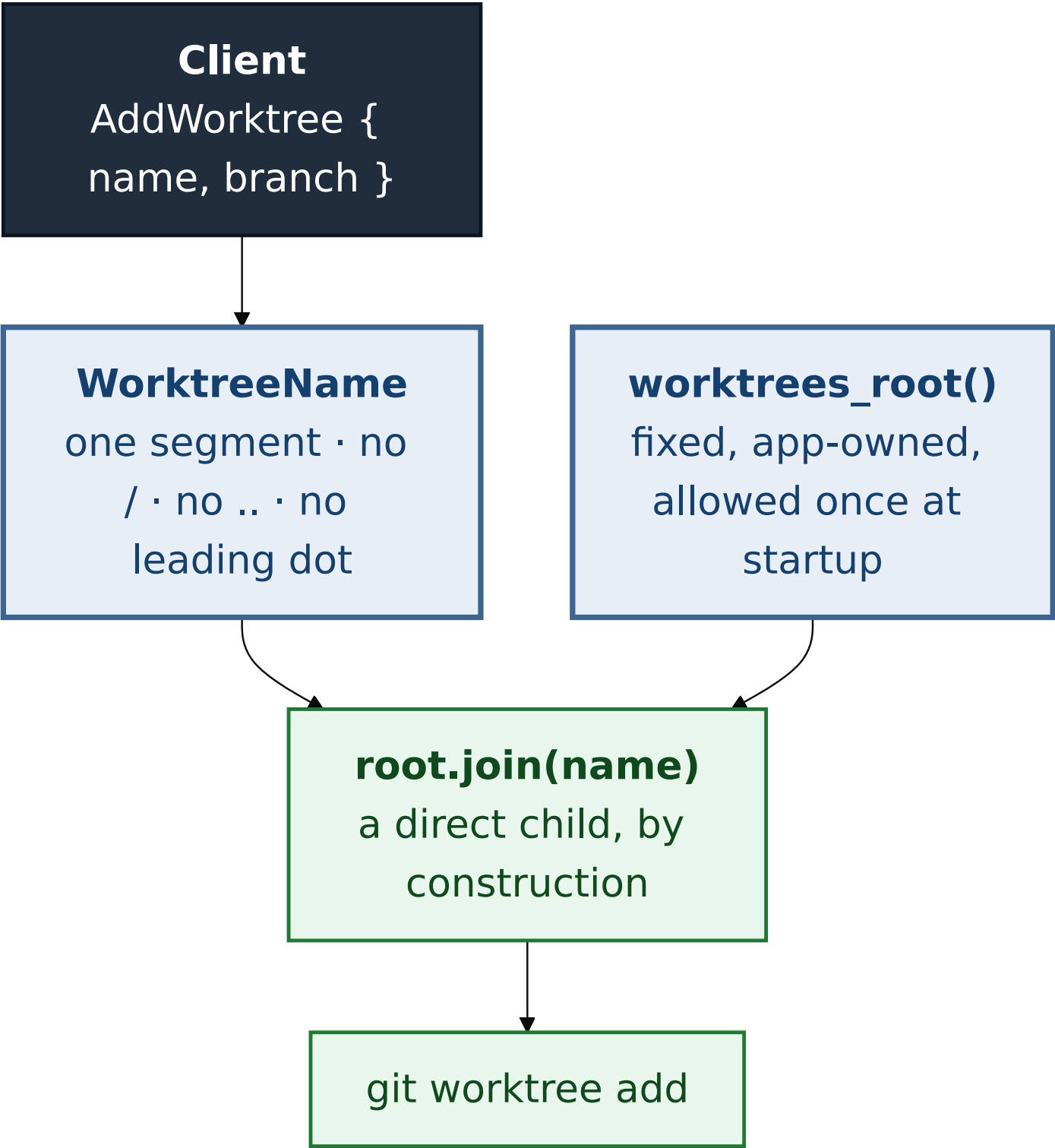
So the operation is:

```
AddWorktree { name: WorktreeName, branch: BranchName },
```

`WorktreeName` is a validated newtype in the protocol crate — **a single path segment**: no separator, no `..`, no leading dot, not empty, length-bounded. The server computes the location itself:

```
worktrees_root().join(name.as_str())
```

A single segment joined to a fixed root can only ever produce a direct child of that root. Containment stops being something the code checks and becomes something the types cannot express otherwise.



3. The load-bearing omission: creating a desk never widens the fence

ADR 0117 established that admitting a *discovered* worktree must not call `allow_root`. The same rule applies here, and the temptation is stronger, because here the directory genuinely is new and "it doesn't work until you allow it" is a plausible-looking bug report.

It already works. The **root** is allowed once, at startup, as a constant of the installation; every child of it is therefore already inside the fence. Adding `allow_root(new_path)` per created worktree would grow the allowlist by one entry per creation, and each entry would outlive the directory it was added for — so a name that ever escaped validation would leave a permanently admitted root behind it.

Allowing the **root** and allowing a **request-derived path** look similar and are not:

| what is allowed                   | when                                  | derived from                     |
|-----------------------------------|---------------------------------------|----------------------------------|
| <code>worktrees_root()</code>     | once, at startup                      | the installation's configuration |
| <code>clones_root()</code>        | once, at startup (and on first clone) | the installation's configuration |
| a created worktree's directory    | <b>never</b>                          | would be a request               |
| a discovered worktree's directory | <b>never</b> (ADR 0117)               | would be a census                |

3a. "Allowed once, at startup" was a sentence with nothing performing it

The row above says `worktrees_root()` is allowed *once, at startup*. As first implemented, **nothing did that**. The root was resolved (`worktrees_root()`) and written to (`create_dir_all` in the executor), and no code path ever admitted it: there was a `scan_clones_root()` at startup and on rescan, and no worktrees equivalent. So the containment argument this whole ADR rests on — *a new desk is inside the fence by construction* — held only as prose, and a desk the app had just created could not be selected (grok, reviewing PR #656).

The correction is `state::scan_worktrees_root()`, called at startup beside `scan_clones_root()` and again on `POST /api/rescan`. It is a **scan**, not a bare `allow_root`, because admitting the root and registering the desks already under it are the same job — exactly as they are for clones, and reusing that shape means one mechanism rather than two that can drift.

Two details are load-bearing rather than incidental:

- `read_only: false`. That flag marks an entry a URL clone (ADR 0008) and is what the picker keys **Delete** on. A linked worktree is an ordinary working tree of a repository the operator already has; marking it a clone would offer to delete it through a route whose guard is *"canonicalizes inside the clones root"*, which this root deliberately does not.

- **create\_dir\_all before the scan.** `Catalog::scan_direct_children` returns early — *before* `allow_root` — when `read_dir` fails. On a fresh install the root does not exist, so without this the admission never happens at all and the first `AddWorktree` produces a directory nothing can serve. The clones scan creates its root for a cosmetic reason (a misworded warning); here it is the mechanism.

**The general lesson, and it is the one worth carrying:** a security argument of the form "*X is admitted once, so every child is safe*" has two halves, and the omission half (§3 — never widen per-path) is the one that gets pinned, because a widening is what reviewers look for. The *admission* half is the one nobody tests, because forgetting it makes the app serve **less**, and a feature that quietly does not work reads as a bug rather than as a fence failure. Both halves now have tests: the exact-body pin for the omission, and

`a_missing_managed_root_is_created_admitted_and_its_desks_are_servable` plus `a_desk_the_app_just_made_is_serviceable_in_the_census` for the admission — the second asserting the user-visible outcome (`Serviceable::Yes` in the census) rather than the mechanism.

**'Admitted once,  
so every  
child is safe'**  
the containment  
argument

```
graph TD; A["'Admitted once, so every child is safe'  
the containment argument"] --> B["Half 1: ADMIT the root  
forgetting it makes the app  
serve LESS"]; A --> C["Half 2: never widen per path  
doing it makes the app  
serve MORE"]; B --> D["Reads as a bug  
'the new desk won't open'  
— so nobody tested it"]; C --> E["Reads as a security hole  
— so it got an exact-body pin"]; D --> F["Both halves now tested  
admission:  
Serviceable::Yes  
in the census  
omission: the exact-body pin"]; E --> F;
```

**Half 1: ADMIT  
the root**  
forgetting it  
makes the app  
serve LESS

**Half 2: never  
widen  
per path**  
doing it makes  
the app  
serve MORE

**Reads as a bug**  
'the new desk  
won't open'  
— so nobody  
tested it

Reads as a  
security hole  
— so it got an  
exact-body pin

**Both halves  
now tested**  
admission:  
Serviceable::Yes  
in the census  
omission: the  
exact-body pin

## 4. Every failure of this operation withholds the path by default

The destination is the one thing the user did not choose and cannot see — §2 is built on that. Three of `exec_add_worktree`'s failure arms handed it to them anyway, the worst being git's own refusal relayed with only `.trim()` applied:

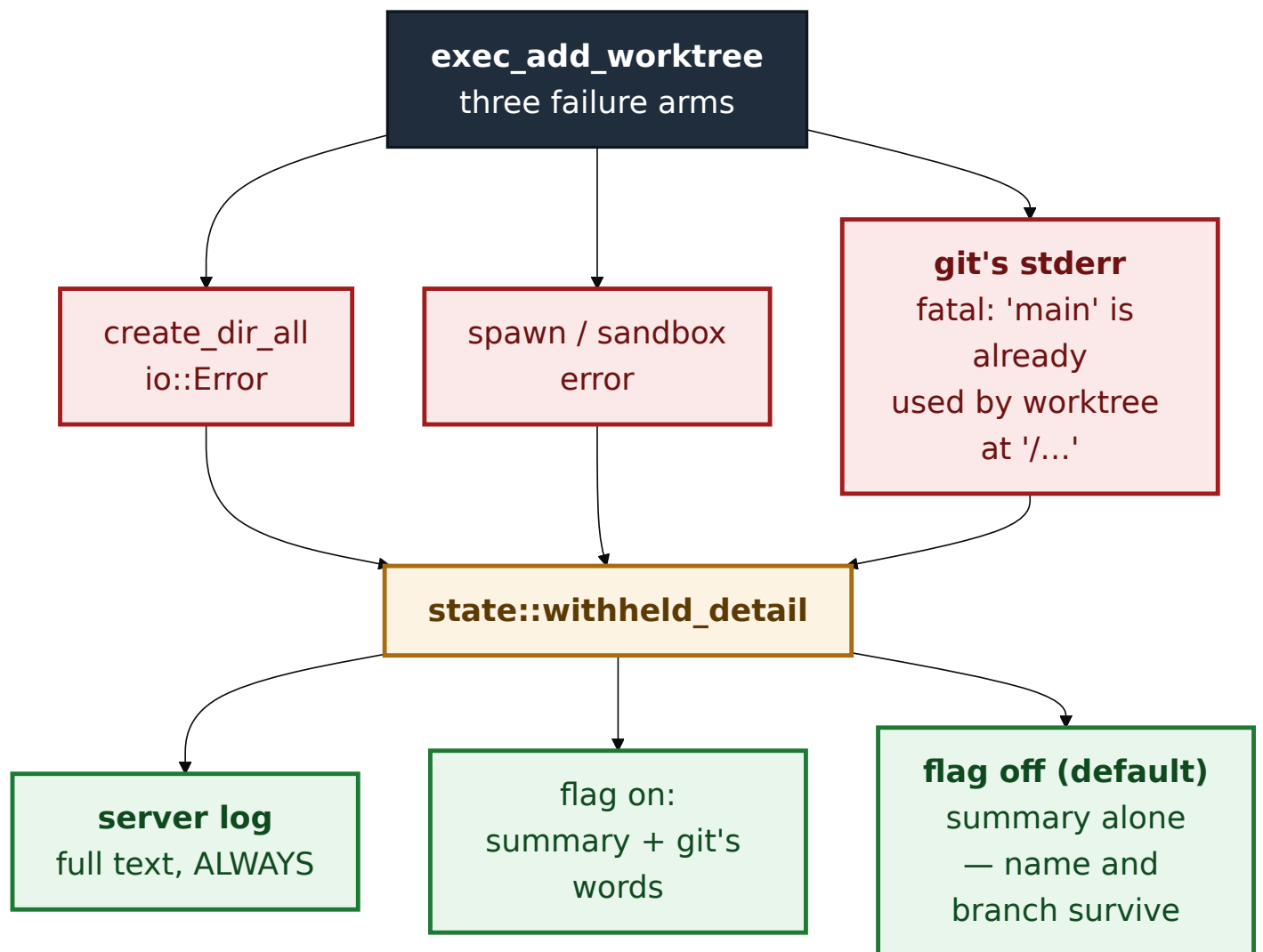
```
fatal: 'main' is already used by worktree at '/home/.../.local/share/git-vist
```

That shipped in the HTTP body regardless of `GIT_VISTA_EXPOSE_PATHS` (reproduced independently by codex and grok on PR #656). It is the same shape as #657 on the worktree census, one milestone over: **a path-exposure control that holds on the success arm and not on the failure arm.** Twice in one milestone is a pattern, not a coincidence — the success arm is what everyone tests.

All three arms now go through `state::withheld_detail`: this function writes the client-safe sentence itself, git's own words are appended **only** when the operator opted in, and the full text is written to the server's log either way. The rule is ADR 0119's — *"A guarantee that holds only on the success arm is not a guarantee"*, which lands with #657 / PR #658, still open at the time of writing, so the link is deliberately omitted rather than left to 404 (the same habit this ADR's own header follows for 0117). It is applied without exception here: a string that arrived from git or from the OS is *detail*, and it is not inspected first to decide.

The refusal stays actionable because the two things the user actually chose — the desk name and the branch — are their own words, so the composed sentence still names both and says what the rule is. What is withheld is only where on disk the collision lives.

**The test that would have stayed green is fixed in the same change.** `a_desk_on_the_branch_you_are_standing_on_is_refused` asserted `status != OK` and that the destination was absent, and both were true throughout the leak. It now reads the body and asserts it against *this run's actual temporary directories*, with a paired positive that the name and branch survived — so redaction cannot pass by making the message useless.



## Alternatives considered

### The sibling-directory convention — declined, but it was a real candidate

Tom's own habit is visible in his machine's worktree list: `Git-Vista-336-wt`, `Git-Vista-codex-379`, `Git-Vista-testbed-8081` — a sibling directory next to the main repo, named for the issue. The arguments for following it are genuine and should not be flattened:

- **It matches what he already does.** A tool that puts things where its user already looks for them is friendlier than one that invents a location.
- **The path is predictable and typeable.** He can `cd` to it without asking the app where it went.
- **It needs no new root** to configure, to back up, or to explain.
- **On his machine it would already be inside the fence**, because `~/projects` is the configured repo root — so a sibling is a child of an allowed root and would be discovered and registered by the existing scan for free. That is a real advantage, and it is exactly the gap ADR 0117 had to build a route to close.



It is declined because **that last advantage is a property of one machine's configuration, not of the design** — and this is measurable rather than hypothetical. The browser harness registers repositories explicitly through `GIT_VISTA_REPOS`, which allows each repository's *own* path as a root and nothing above it. #548's fixture places a worktree beside its repository under exactly that configuration, and the census resolves it

`Serviceable::OutsideAllowedRoots` — asserted in CI, in `worktree-drawer.spec.mjs`, where the app shows the fence sentence and refuses to open it.

So the sibling convention produces a worktree the app cannot open, on an ordinary supported configuration, and the app would have created it itself. The fix would be to check containment at every path-picking site and fall back when it fails — which is precisely the "checked every time" rule Tom's answer names as the thing that holds until one code path forgets it.

**What is lost, stated plainly:** the created worktree will not be where Tom's muscle memory expects, and this ADR does not pretend that costs nothing. Two things soften it and neither is a promise this ADR makes on behalf of a later one: the drawer (#548) shows every worktree with its branch, and shows the absolute path when `GIT_VISTA_EXPOSE_PATHS` is set; and nothing here stops a future issue from adding an operator-configured root, since the location is already one function.

## **A client-supplied path validated against the root — declined**

The shape the spec sketched, with a containment check on arrival. It is not unsafe if the check is right; it is declined because it makes the safety a property of a check rather than of the construction, and because it would be the only request body in this codebase that carries a filesystem path. The failure mode is the ordinary one for canonicalisation checks — symlinks, `..` in an encoding the validator did not consider, TOCTOU between check and use — and the named form has none of them to get wrong.

## Not building `AddWorktree` at all — considered, and rejected as premature to reject

The spec's open question 1 notes that M11.01-03 delivers most of the value with no path policy at all, and that closing this as "not planned" is legitimate. That remains true. It is not taken, because the decision that blocked it has now been made and the remaining work is small — but the argument is recorded so a later reader knows it was weighed rather than overlooked.

## Consequences

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- One new managed root appears on disk. Like the clones root it is created lazily and is empty on a fresh install.
- `AddWorktree` is `RiskLevel::Safe` with `RecoveryStrategy::NotNeeded`: it creates a directory and a metadata file, moves no ref, and destroys nothing. git itself refuses if the branch is already checked out elsewhere — the precondition ADR 0116 added states that rule to the UI first.
- Worktrees created here are discovered by the census like any other, and resolve `Serviceable::Yes` because the root is allowed — so #548's drawer lists and opens them with no new code.
- The three effect classifiers in `git-vista-protocol/src/effects.rs` gain an arm each (the compiler forces this, ADR 0091), and `tests/explain_parity.rs`'s independent table gains its row. Both are required: the parity table is what keeps the derived half from being unpinned.
- **A `WorktreeName` that fails validation is refused before any path exists.** The refusal is the server's, not the picker's, and is tested as such.
- **No UI ships with this slice, deliberately.** The natural home for a "new desk" control is #548's drawer, which is still in review — building a second one here would mean duplicating it and then deleting one. This follows the staging M11.01 already used: the primitive and its wire type land and are reviewed first, with no route into the UI yet. The browser suite is therefore unchanged by this slice, and its number is stated in the PR as evidence of that rather than as evidence of new coverage.